APPENDIX B. USER'S GUIDE

USER'S GUIDE

AN INTELLIGENT MACHINE LEARNING SYSTEM FOR PROGRAMMING DIFFICULTY ASSESSMENT AND CODING ENGAGEMENT ENHANCEMENT

1. Introduction

This User's Guide provides step-by-step instructions for using the **An Intelligent Machine Learning System for Programming Difficulty Assessment and Coding Engagement Enhancement**. The system is developed to assist in managing student information, assessing programming code, identifying programming languages, detecting coding errors or correctness, generating recommendations, and viewing assessment reports.

The system consists of three major features:

1) **Student Management**
2) **Coding Assessment**
3) **Report**

This guide is intended for authorized users, instructors, administrators, and students who will interact with the system.

2. System Access

To access the system, the user must have a compatible device and a supported web browser.

Step 1: Open the Browser

Open a web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge.

Step 2: Access the System

Enter the system's assigned web address in the browser's address bar.

Step 3: Login

Enter the required account credentials:

- Username
- Password

Click the **Login** button.

If the credentials are valid, the system will display the user's dashboard.

3. Dashboard

After successful login, the system displays the main dashboard.

The dashboard provides access to the major functions of the system:

- Student Management
- Coding Assessment
- Report
- Logout

- The dashboard serves as the main navigation area of the system.

4. Student Management

The **Student Management** feature allows authorized users to manage student information.

Users can:

- Add student records
- View student records
- Search student records
- Edit student information
- Delete student records

4.1 Adding a student

To add a new student:

1. Click **Student Management**.
2. Click **Add Student**.
3. Enter the required student information.
4. Review the information.
5. Click **Save**.
6. The system stores the student record.
7. The new student will appear in the student list.

The information may include:

- Student ID
- Full Name
- Gender
- Year Level
- Program/Course
- Contact Information
- Other required information

4.2 Viewing Student Information

To view a student's information:

1. Open **Student Management**.
2. Locate the desired student.
3. Click **View**.
4. The student's information will be displayed.

4.3 Searching for a Student

To search for a student:

1. Open **Student Management**.
2. Locate the search field.
3. Enter the student's name or ID number.
4. Click **Search**, if required.

5. The system displays matching student records.

4.4 Editing Student Information

To update a student record:

1. Locate the student.
2. Click **Edit**.
3. Modify the required information.
4. Review the changes.
5. Click **Update** or **Save**.
6. The system updates the student record.

4.5 Deleting a student

To delete a student record:

1. Locate the student record.
2. Click **Delete**.
3. Review the selected record.
4. Confirm the deletion.
5. The system removes the selected record.

Reminder: Users should verify the selected student before confirming deletion.

5. Coding Assessment

The **Coding Assessment** is the main intelligent feature of the system.

It allows the user to submit source code and have the system analyze it.

The system is designed to:

1. Identify the programming language used.
2. Analyze the submitted code.
3. Determine whether the code is correct or contains errors.
4. Identify possible coding difficulties.
5. Generate recommendations.
6. Record the assessment result.

6. Starting a Coding Assessment

To perform a coding assessment:

1. Open **Coding Assessment**.
2. Select the student to be assessed.
3. Enter or paste the source code into the coding area.
4. Review the submitted code.
5. Click **Analyze Code** or the corresponding assessment button.
6. Wait while the system processes the code.
7. Review the assessment results.

7. Programming Language Detection

The system analyzes the submitted source code to determine the programming language being used.

Depending on the programming languages supported by the implemented system, the result may identify languages such as:

- HTML
- CSS
- JavaScript
- Python
- Java
- C++
- Other supported languages

Procedure

1. Enter the source code.
2. Click **Analyze Code**.
3. Wait for the analysis.
4. Check the **Detected Programming Language** result.

The detected language is based on recognizable characteristics or syntax within the submitted code.

8. Code Correctness and Error Detection

After identifying the programming language, the system analyzes the submitted code.

The system may classify the submitted code as:

- **Correct**
- **Contains Error**
- **Needs Improvement**

Procedure

1. Submit the source code.
2. Start the assessment.
3. Wait for the analysis to finish.
4. Review the code status.
5. Examine the detected error or problem, if available.
6. Read the recommendation generated by the system.

The purpose of this feature is to provide useful feedback that can help students understand and improve their coding performance.

9. Viewing the Recommendation

After analyzing the source code, the system generates a recommendation based on the assessment.

The recommendation may guide the student to:

- Review programming syntax.
- Check the identified error.
- Review a particular programming concept.
- Correct a missing or incorrect element.

- Improve the programming logic.
- Practice the related programming topic.
- Resubmit the corrected code.

Procedure

1. Complete the coding assessment.
2. Locate the **Recommendation** section.
3. Read the generated recommendation.
4. Apply the suggested correction.
5. Modify the source code.
6. Submit the code again for reassessment.

10. Reassessment

The system allows the student to improve the submitted code and perform another assessment.

Procedure

1. Review the previous assessment.
2. Identify the detected problem.
3. Follow the generated recommendation.
4. Correct the source code.
5. Submit the revised code.
6. Run the assessment again.

7. Review the new result.

This process allows students to practice coding and learn from identified errors.

11. Report

The **Report** feature provides organized information about coding assessments.

It allows authorized users to review assessment records and monitor students' coding performance.

The report may contain:

- Student Name
- Student ID
- Assessment Date
- Programming Language
- Submitted Code
- Code Status
- Detected Error
- Recommendation
- Assessment Result

12. Viewing Reports

To view an assessment report:

1. Click **Report**.

2. Select the desired student or assessment record.
3. Apply available filters if necessary.
4. Click **View**.
5. Review the displayed assessment information.

13. Searching Reports

To search for a particular assessment record:

1. Open the **Report** module.
2. Locate the search field.
3. Enter the student's name or ID.
4. Click **Search**.
5. Review the matching records.

If available, reports may also be filtered according to:

- Student
- Year Level
- Programming Language
- Assessment Date
- Assessment Status

14. Monitoring Coding Performance

The Report module can be used by authorized users to review students' coding assessment records.

The instructor or administrator can review:

1. Completed assessments.
2. Programming languages detected.
3. Correct code submissions.
4. Code submissions containing errors.
5. Identified coding difficulties.
6. Generated recommendations.
7. Assessment history.

These records can provide useful information for monitoring students' programming activities and identifying areas that may require additional instruction or practice.

15. Logout

After completing the activities, the user should log out of the system.

Procedure

1. Locate the **Logout** button.
2. Click **Logout**.
3. The system ends the current session.
4. The user is returned to the login page or system landing page.

Users should always log out when using a shared computer.